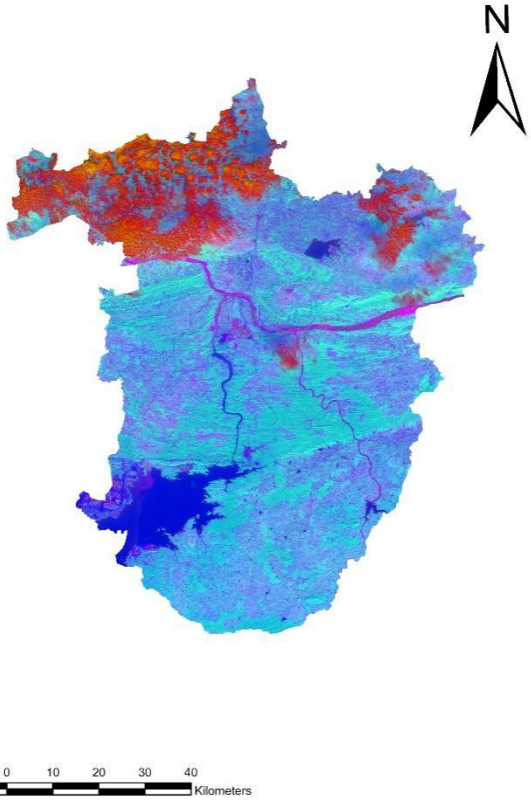
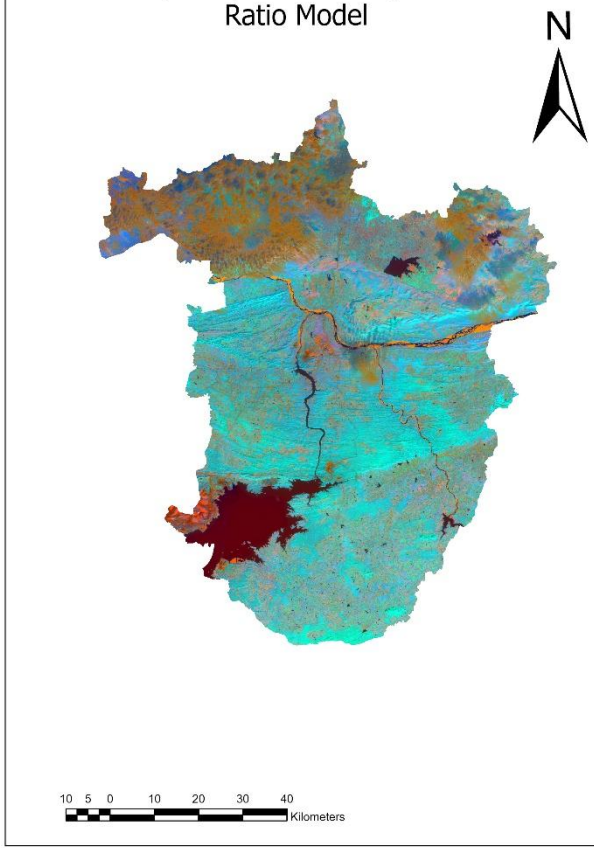
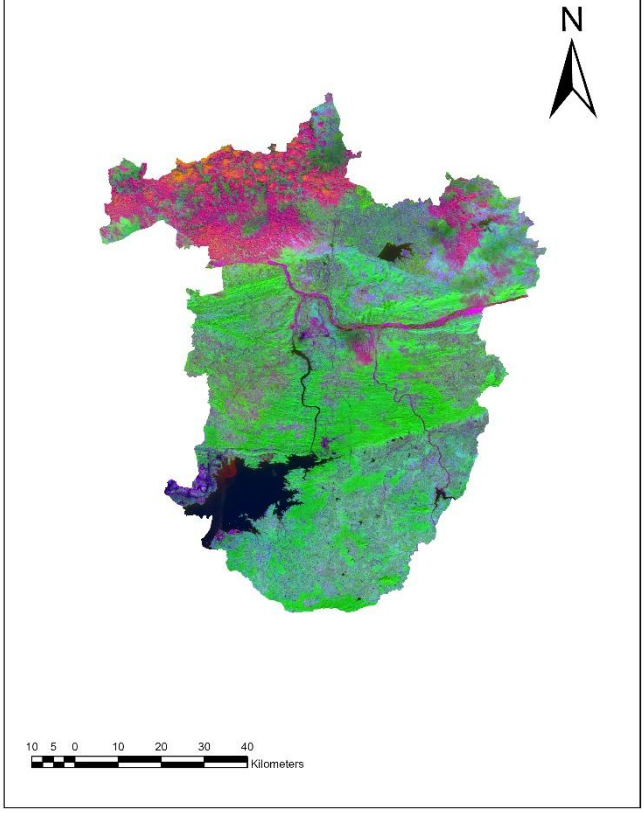
Clay Mineral Mapping of Sonbhadra District, Uttarpradesh using Hashim Pour Ratio Model



Clay Mineral Mapping of Sonbhadra District ,Uttarpradesh using Kaufmann's Ratio Model



Clay Mineral Mapping of Sonbhadra district, Uttarpradesh using Sabins' Ratio Model



5.COLOUR INTERPRETATION OF SABIN RATIO MODEL:

The clay mineral map of Sonbhadra district using the Sabins' Ratio Model shows clear mineral differentiation through colour variations. Magenta to pink shades indicate clay minerals such as kaolinite, illite, and montmorillonite, mainly concentrated in the northern region. Green tones represent dense vegetation, masking underlying geology. Reddish to orange areas suggest iron oxides like hematite and limonite, associated with weathered surfaces. Light cyan shades indicate mixed soil and sparse vegetation zones. Dark blue to black regions correspond to water bodies. Overall, the model effectively highlights clay-rich zones and weathered mineral surfaces.

COLOUR INTERPRETATION OF KAUFMANN RATIO MODEL:

The clay mineral map of Sonbhadra district using the Kaufmann's Ratio Model shows clear mineral distribution through colour variations. Cyan to light blue shades indicate clay minerals such as kaolinite, illite, and montmorillonite. Brown to orange tones represent iron oxides like hematite and limonite, suggesting weathered or oxidized zones. Turquoise shades may indicate alteration minerals such as chlorite and epidote, while purple patches suggest sericite/muscovite presence. Dark red to black areas correspond to water bodies. Overall, the map effectively highlights clay-rich and alteration zones useful for mineral exploration.

COLOUR INTERPRETATION OF HASHIM PUR RATIO MODEL:

The clay mineral map of Sonbhadra district using the Hashim Pour Ratio Model highlights hydrothermal alteration and mineral distribution through distinct colours. Red to orange shades indicate strong alteration zones with minerals like alunite, kaolinite, and iron oxides (hematite, limonite). Purple to magenta tones suggest sericite and muscovite, associated with advanced argillic alteration. Cyan to blue areas represent clay minerals such as illite and montmorillonite along with less altered surfaces. Dark blue to black regions correspond to water bodies. Overall, this model effectively identifies hydrothermal alteration zones and clay mineral-rich areas, useful for mineral exploration.

6.Conclusion

- Band ratio techniques successfully identified hydrothermal alteration zones and clay minerals in the study area.
- Sabins model provided simple and effective enhancement, Kaufmann model improved lithological interpretation, and Hashim & Pour model enabled detailed mineral discrimination.
- The integration of all three models improved accuracy and reliability of results, demonstrating the usefulness of remote sensing in mineral exploration.